MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE CDD

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic monostable multivibrator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.9 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CDD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/121-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

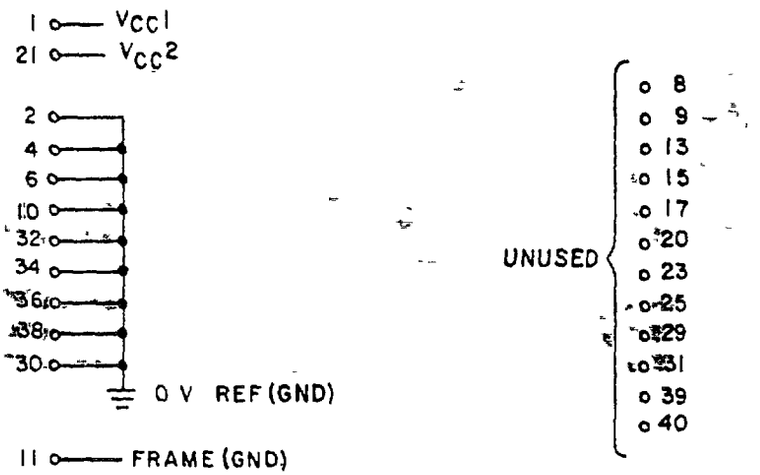
3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

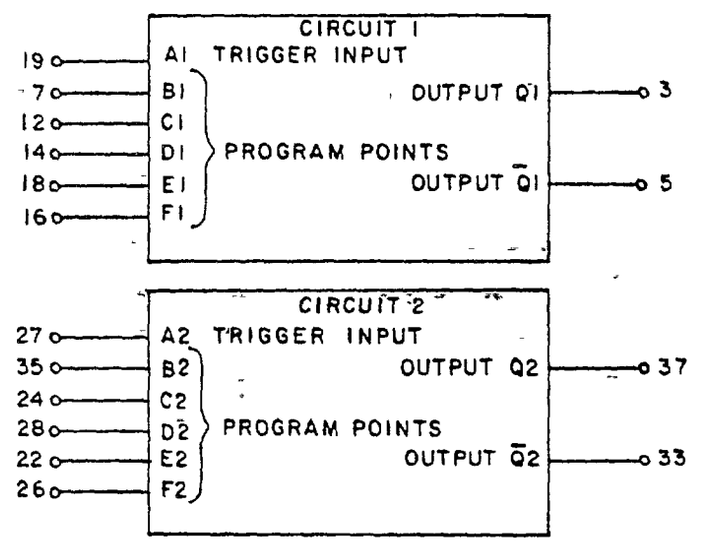
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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* FIGURE 1. CDD module function diagram, monostable multivibrator.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life.		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A7	Continuity	II	X	0	1	0	1	0	1	0	1	Ω
A8	Continuity	II	S	0	1	0	1	0	1	0	1	Ω
B1,B2	Isolation	II	X	10	∞	10	∞	10	∞	10	∞	MΩ
C1,C4, C5,C8	Output voltage, low level	III	X	0	400	0	450	0	400	0	450	mV
C2,C3,C6, C7	Output voltage, high level	III	X	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
D1-D4	Output short circuit current	IV	X	16	65	16	65	16	65	16	65	mA
D5, D7	Input current, high level	IV	X	0	4	0	8	0	6	0	10	µA
D6, D8	Input current, low level	IV	X	0.0	3.2	0.0	3.2	0.0	3.2	0.0	3.2	mA
E1	Power supply current (quiescent)	V	S	0	120	0	120	0	120	0	120	mA
E2	Power supply current (operating)	V	S	0	120	0	120	0	120	0	120	mA
F1-F4	Output pulse width (1.0 µs)	VI	X	0.93	1.07	0.91	1.09	0.90	1.10	0.88	1.12	µs
F5-F8	Output pulse width (2.0 µs)	VI	X	1.86	2.14	1.82	2.18	1.80	2.20	1.76	2.24	µs
F9-F12	Output pulse width (3.0 µs)	VI	S	2.79	3.21	2.73	3.27	2.70	3.30	2.64	3.36	µs
F13-F16	Output pulse width (4.0 µs)	VI	X	3.72	4.28	3.64	4.36	3.60	4.40	3.52	4.48	µs

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
F17-F20	Output pulse width (5.0 µs)	VI	S	4.65	5.35	4.55	5.45	4.50	5.50	4.40	5.60	µs
F21-F24	Output pulse width (6.0 µs)	VI	S	5.58	6.42	5.46	6.54	5.40	6.60	5.28	6.72	µs
F25-F28	Output pulse width (7.0 µs)	VI	S	6.51	7.49	6.37	7.63	6.30	7.70	6.16	7.84	µs
F29-F32	Output pulse width (8.0 µs)	VI	X	7.44	8.56	7.28	8.72	7.20	8.80	7.04	8.96	µs
F33-F36	Output pulse width (9.0 µs)	VI	S	8.37	9.63	8.19	9.81	8.10	9.90	7.92	10.08	µs
F37-F40	Output pulse width (10.0 µs)	VI	S	9.30	10.7	9.10	10.9	9.0	11.0	8.80	11.2	µs
F41-F44	Output pulse width (11.0 µs)	VI	S	10.23	11.77	10.01	12.00	9.90	12.10	9.68	12.32	µs
F45-F48	Output pulse width (12.0 µs)	VI	S	11.16	12.84	10.92	13.08	10.8	13.2	10.56	13.44	µs
F49-F52	Output pulse width (13.0 µs)	VI	S	12.09	13.91	11.83	14.17	11.7	14.3	11.44	14.56	µs
F53-F56	Output pulse width (14.0 µs)	VI	S	13.02	14.98	12.74	15.26	12.6	15.4	12.32	15.68	µs
F57-F60	Output pulse width (15.0 µs)	VI	X	13.95	16.05	13.65	16.35	13.5	16.5	13.2	16.8	µs
G1-G5	Transition time (t_{THL})	VII	S	3	40	3	45	3	40	3	45	ns
G6-G10	Transition time (t_{TLH})	VII	S	10	250	10	250	10	250	10	250	ns
H1,H3,H5	Propagation delay time (t_{PLH})	VIII	S	10	170	10	170	10	170	10	170	ns
H2,H4,H6	Propagation delay time (t_{PHL})	VIII	S	7	170	7	170	7	170	7	170	ns
J1-J4	Output pulse amplitude, low level	IX	S	0	400	0	450	0	400	0	450	mV
J5-J8	Output pulse amplitude, high level	IX	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K1-K4	Output pulse, overshoot	X	S	0	500	0	500	0	500	0	500	mV
K5-K8	Output pulse, undershoot	X	S	0	500	0	500	0	500	0	500	mV
K9-K12	Output pulse, preshoot	X	S	0	500	0	500	0	500	0	500	mV
L1,L3	Output pulse width (recovery time)	XI	S	0.93	1.07	0.91	1.09	0.90	1.10	0.88	1.12	µs
L2,L4	Output pulse width (recovery time)	XI	S	13.95	16.05	13.65	16.35	13.50	16.50	13.20	16.80	µs
M1-M4	Input trigger sensitivity	XII	S	0.93	1.07	0.91	1.09	0.90	1.10	0.88	1.12	µs
M5-M8	Input immunity (20 ns)	XII	S	0	5	0	5	0	5	0	5	ns
M9-M12	Input immunity (0.95 V)	XII	S	0	5	0	5	0	5	0	5	ns
N1,N2	Power supply transient rise time	XIII	S	9	53	9	53	9	53	9	53	µs
N3,N4	Power supply transient rise time	XIII	S	1.5	40.0	1.5	40.0	1.5	40.0	1.5	40.0	µs
N5-N8	Power supply transient false trigger	XIII	S	2.0	5.5	2.0	5.5	2.0	5.5	2.0	5.5	V
P1-P12	Crosstalk	XIV	S	0	5	0	5	0	5	0	5	ns
Q1,Q2	Zener applied potential	XV	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
Q3,Q4	Trigger voltage high	XV	S	0	400	0	450	0	400	0	450	mV
R1,R2	V_{OUT} (low loaded)	XVI	S	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

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4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test condition Q or R is part of a sequence of tests, the order of testing shall be as follows: Q, R where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A,C,D,E, F1-F16, F29-F36, F53-F56, G, H, M1-M4, and N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A and C through N shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through P shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

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* 6.3 Cross reference of classification. Military Specification MIL-M-28787/121 supersedes Department of the Navy, Naval Ordinance Systems Command Drawing 10001-2658710. Part number M28787/121-1 supersedes part numbers 2658710 and M28787/121.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	5.0 V dc ±10Δ
Maximum power dissipation	660 mW (outputs not loaded)
Input voltage	
Low level	0.0 V minimum, 0.95 V maximum
High level	3.2 V minimum, V_{CC} maximum
Input current	
Low level	3.2 mA minimum
High level	10.0 µA maximum
Output voltage	
Low level	0.45 V maximum
High level	3.5 V minimum
Q output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	35.2 mA
High level (V_{OH} = 3.8 V)	200 µA
Q output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	22.4 mA
High level (V_{OH} = 3.8 V)	200 µA
Input trigger pulse width	85 ns minimum
Output pulse width range	0.88 µs minimum, 16.8 µs maximum
Maximum recovery time	4.0 µs

* 6.5 Module function. The module consists of two monostable multivibrator (one-shot) circuits and a range of discrete resistors and capacitors. Each one-shot circuit is externally pin programmable so as to deliver output pulse widths in the range of 1 to 15 microseconds in 1 microsecond increments. For each input trigger pulse, each circuit delivers a positive output pulse Q and its complement pulse $\overline{Q}$ simultaneously. Each circuit is triggered on the trailing edge of a positive input pulse. Each circuit is capable of being driven by a source with an impedance of 12 kilohms or less when the input voltage level is high and an impedance of 100 ohms or less when the input voltage level is low. The $\overline{Q}$ output is capable of driving fourteen standard logic loads (22.4 mA) at a low level. The Q output is capable of driving twenty-two standard logic loads (35.2 mA) at a low level. Each output is capable of driving forty standard logic loads (200 µA) at a high level.

6.5.1 Input to output relationship. Each monostable multivibrator circuit has the input to output relationship shown on figure 1.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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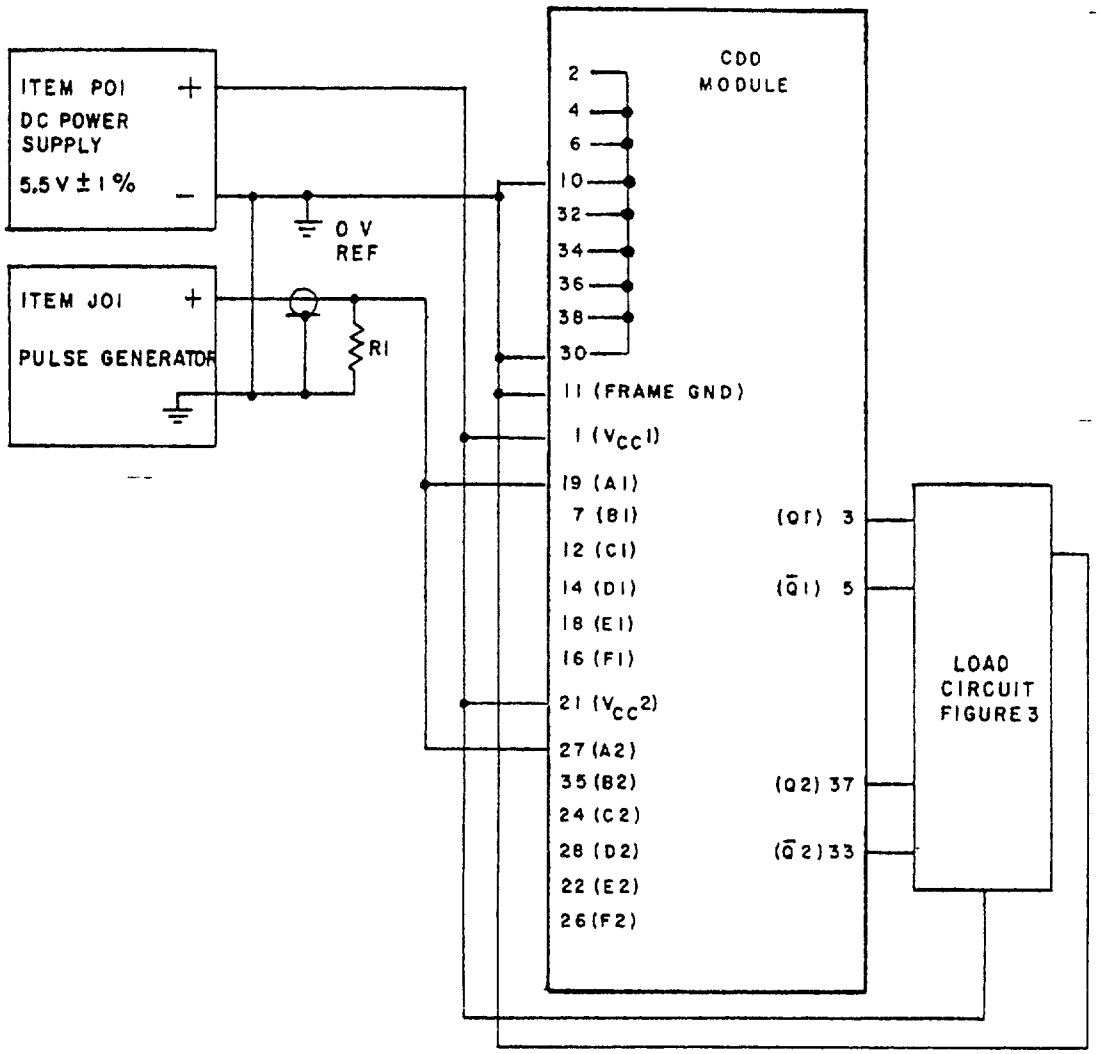
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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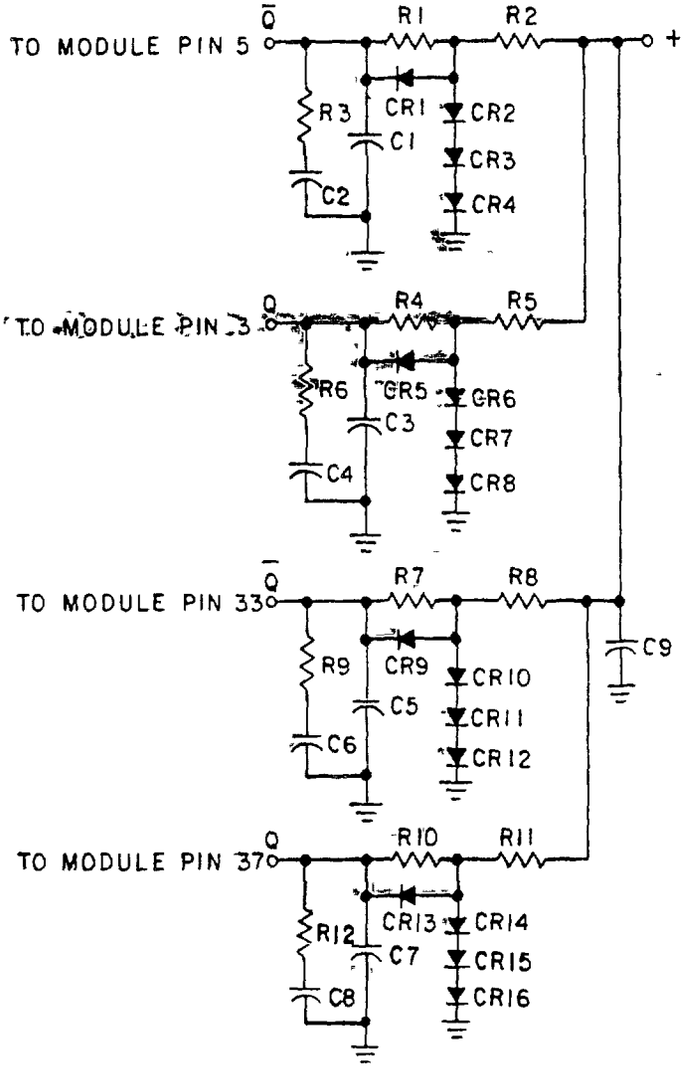
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Item J01 parameters:
   PA = 5 V ±5%.
   PW (1.5 V level) = 150 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 30 µs ±10%.

* FIGURE 2. Operating temperature, component temperature, and life test circuit.

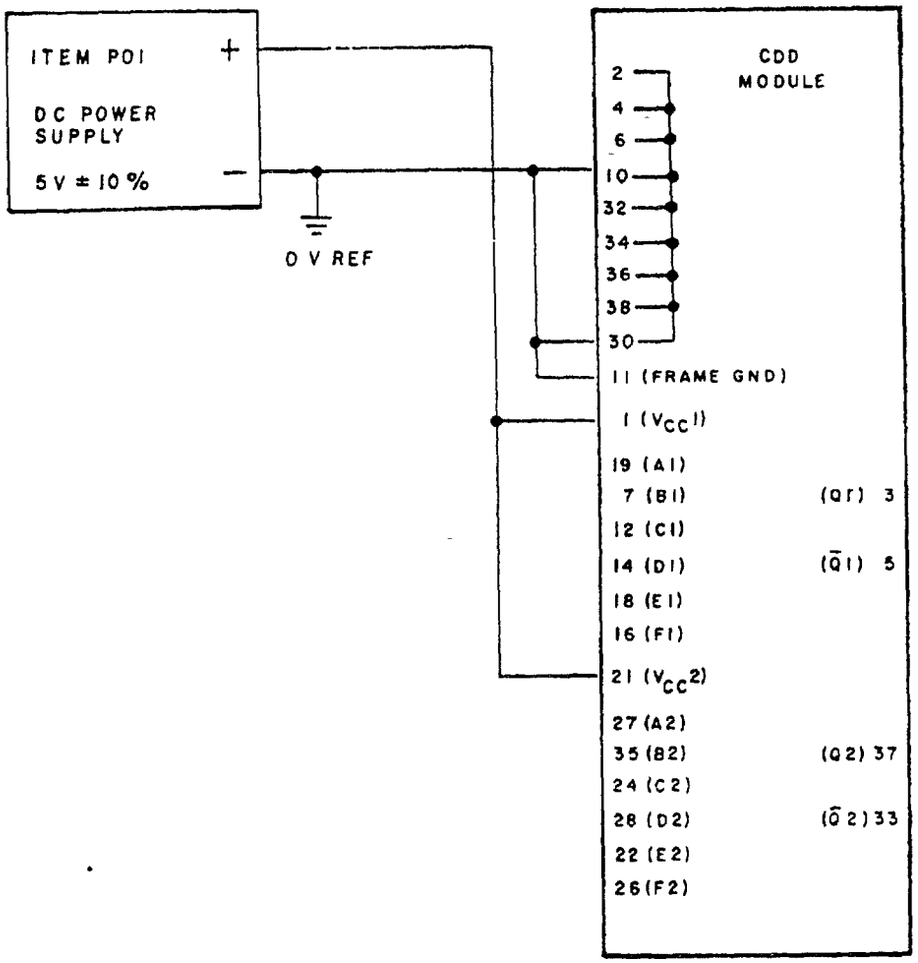
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NOTES:
1. R1 = R7 = 7.5 kΩ ±1%, 1/8 W.
2. R2 = R8 = 215Ω ±1%, 1/8 W.
3. R3 = R6 = R9 = R12 = 100Ω ±1%, 1/8 W.
4. R4 = R10 = 7.15 kΩ ±1%, 1/8 W.
5. R5 = R11 = 140Ω ±1%, 1/8 W.
6. CR1-CR16 = 1N4150, or equivalent.
7. C1 = C3 = C5 = C7 = 33 pF ±10%.
8. C2 = C4 = C6 = C8 = 330 pF ±10%.
9. C9 = 1.0 µF ±10%, 50 V dc.
10. Loads shall be assembled on a board less than 3 by 3 (76.2 by 76.2 mm) with components and wiring within 1/2 inch (12.7 mm) of the board.
11. The maximum lead length from load to the appropriate module pin shall be 2 inches (50.8 mm).

FIGURE 3. CDD module load circuit.

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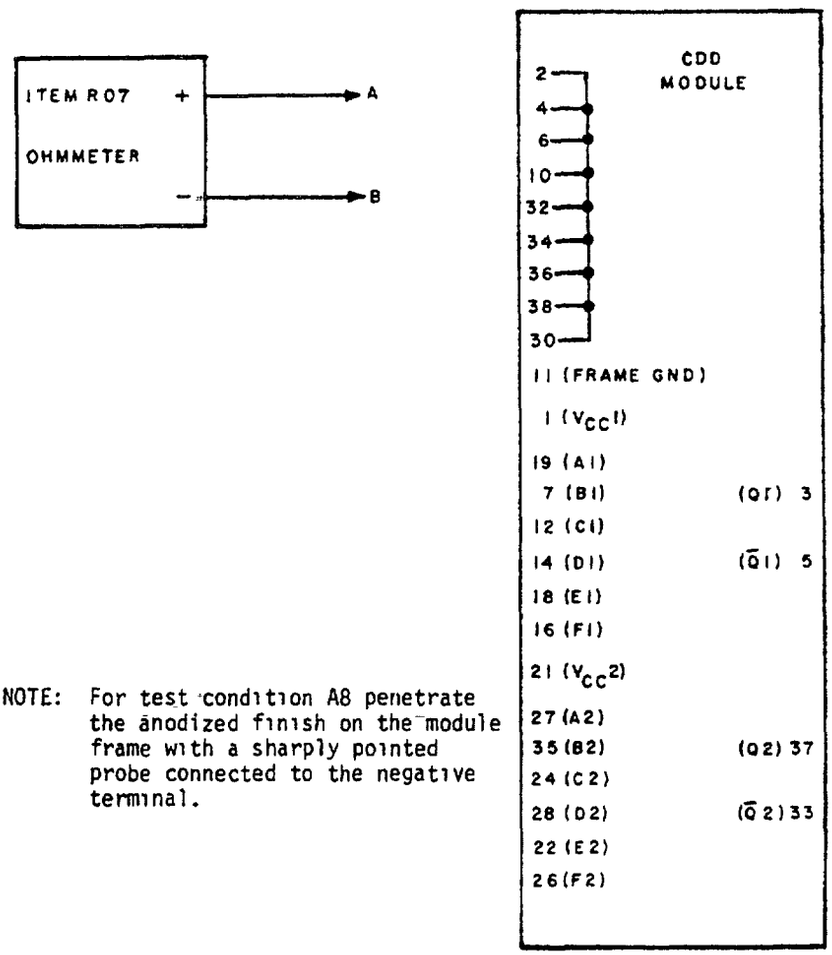
* FIGURE 4. Vibration test circuit.

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* TABLE II. Continuity and isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Continuity	10	2	0	1	Ω	5
A2	↓	10	4	↓	↓	↓	↓
A3		10	6				
A4		30	32				
A5		30	34				
A6		30	36				
A7		30	38				
A8	↓	11	Frame	↓	↓	↓	
B1	Isolation	11	All other pins interconnected	10	∞	MΩ	
B2	Isolation	Unused pins 8,9,13,15, 17,20,23, 25,29,31, 39,40, interconnected	To all other pins interconnected	10	∞	mΩ	↓

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NOTE: For test condition A8 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

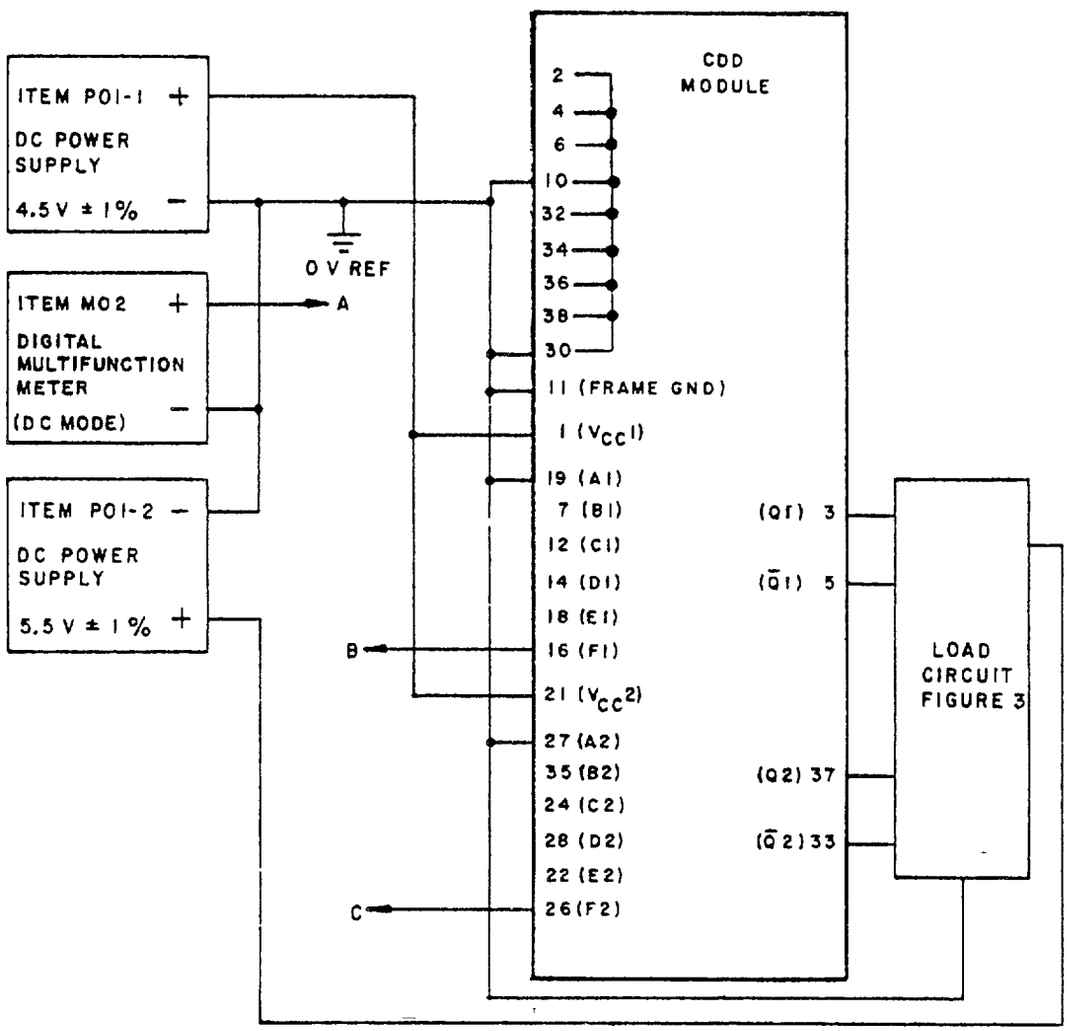
* FIGURE 5. Continuity and isolation test circuit for table II.

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*TABLE III. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	Q1 Output voltage, low level	3	Open	Open	0	400	mV	6
C2	$\overline{Q1}$ Output voltage, high level	5	Open	↓	3.8	5.5	V	↓
C3	Q1 Output voltage, high level	3	10	↓	3.8	5.5	V	↓
C4	$\overline{Q1}$ Output voltage, low level	5	10	↓	0	400	mV	↓
C5	Q2 Output voltage, low level	37	Open	↓	0	400	mV	↓
C6	$\overline{Q2}$ Output voltage, high level	33	↓	↓	3.8	5.5	V	↓
C7	Q2 Output voltage, high level	37	↓	30	3.8	5.5	V	↓
C8	$\overline{Q2}$ Output voltage, low level	33	↓	30	0	400	mV	↓

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NOTES:

1. Prior to making measurement for test conditions C3, C4, C7 and C8, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. Interconnecting conductor length between pins 10 and 16 and between pins 26 and 30 shall not exceed 6 inches (152.4 mm).

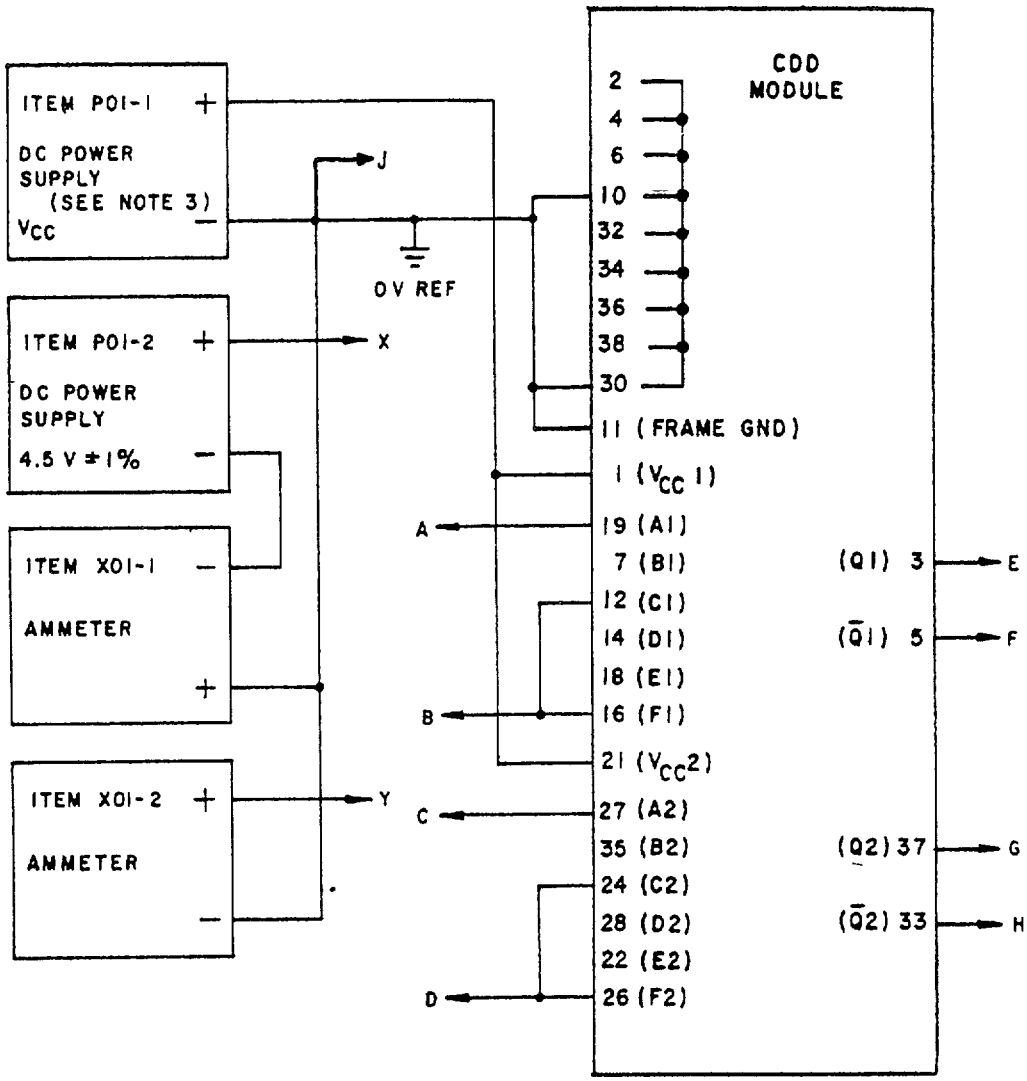
* FIGURE 6. Output voltage (low-level and high-level) test circuit for table III.

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* TABLE IV. Output short circuit current and input current (high-level and low-level) test.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
D1	Q1 Output short circuit current	J	10	J	Open	Y	Open	Open	Open	16	65	mA	7
D2	$\overline{Q1}$		Open		Open	Open	Y	Open	Open				
D3	Q2				30		Open	Y	Open				
D4	$\overline{Q2}$				Open			Open	Y				
D5	C1 and C2 Input current, high level	X		Open					Open	0	4	µA	
D6	C1 and C2 Input current, low level	Y		Open						0.0	3.2	mA	
D7	C1 and C2 Input current, high level	Open		X						0	4	µA	
D8	C1 and C1 Input current, low level	Open		Y						0.0	3.2	mA	

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CAUTION:

The chassis of power supply P01-2 shall be isolated from earth ground.

CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:

1. Prior to making measurements for test conditions D1 and D3, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. The conductor lengths between points B and D, and their respective columns shall not exceed 6 inches (152.4 mm).
3. V_{CC} = 5.5 V ±1% for all test conditions except for test conditions D5 and D7 where V_{CC} = 4.5 V ±1%.

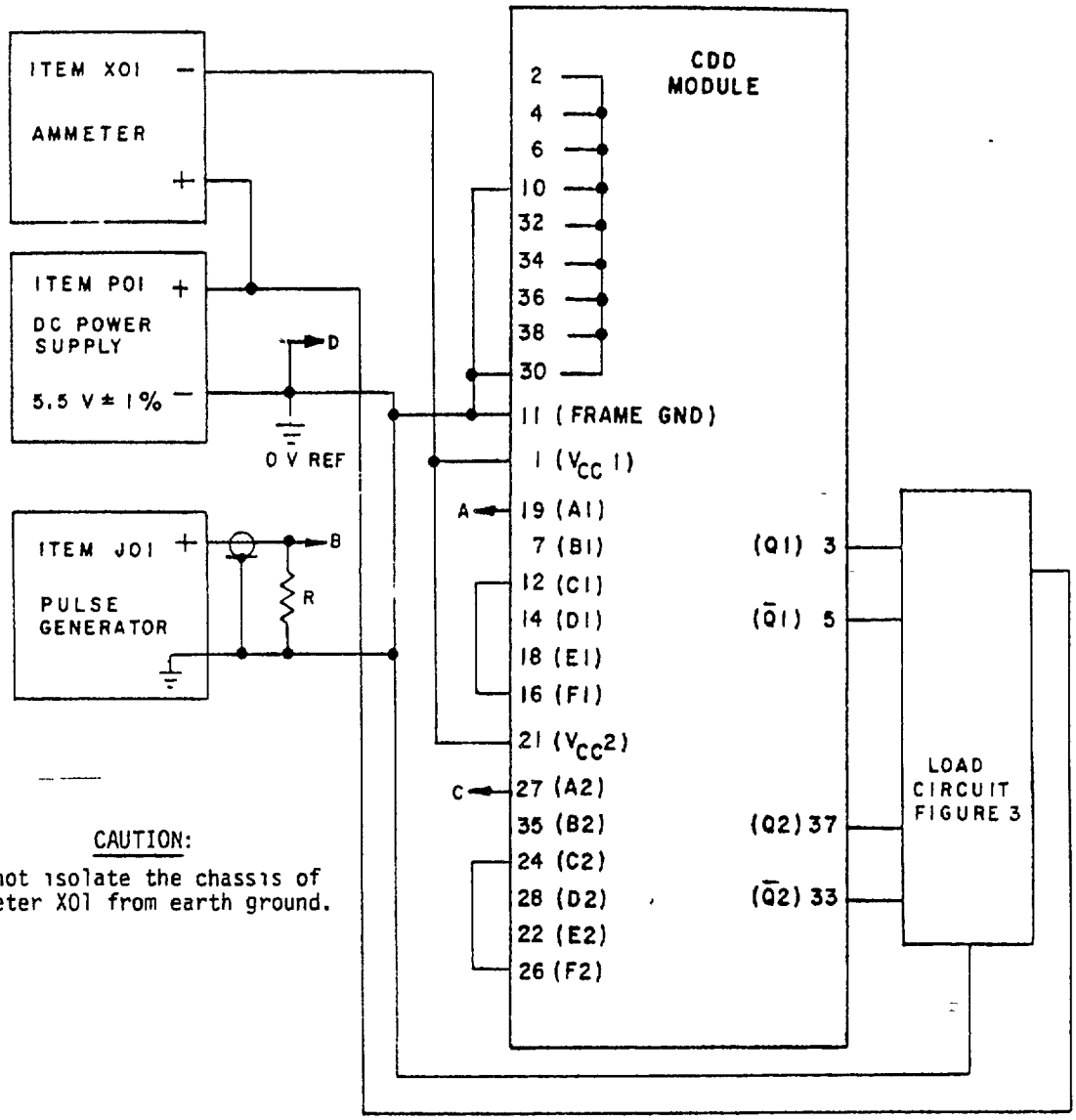
* FIGURE 7. Output short circuit current and input current (high-level and low-level) test circuit for table IV.

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* TABLE V. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	C to pin	Min	Max		
E1	Power supply current (quiescent)	D	D	0	120	mA	8
E2	Power supply current (operating)	B	B	0	120	mA	8

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NOTES:
1. R = 50Ω ±5%, 1 W.
2. Input pulse parameters:

PA = 4 V ±5% with ref to 0 V.
PW (1.5 V level) = 125 ns ±10%.
$t_{TLH} = t_{THL}$ = 20 ns ±10%.
PRP = 5.0 µs ±10%

* FIGURE 8. Power supply current test circuit for table V.

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* TABLE VI. Output pulse width test.

Test condition	Parameter	V_{CC}	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F1	Q1 Output pulse width (1.0 µs)	5.5	12	16	Open	Open	19	3	0.93	1.07	µs	9,10
F2	$\overline{Q1}$	4.5	12	16			19	5				
F3	Q2	4.5	24	26			27	37				
F4	$\overline{Q2}$	5.5	24	26			27	33				
F5	Q1 Output pulse width (2.0 µs)	5.5	7	12	14	16	19	3	1.86	2.14		
F6	$\overline{Q1}$	4.5	7	12	14	16	19	5				
F7	Q2	4.5	24	35	26	28	27	37				
F8	$\overline{Q2}$	5.5	24	35	26	28	27	33				
F9	Q1 Output pulse width (3.0 µs)	4.5	14	16	Open	Open	19	3	2.79	3.21		
F10	$\overline{Q1}$	5.5	14	16			19	5				
F11	Q2	5.5	26	28			27	37				
F12	$\overline{Q2}$	4.5	26	28			27	33				
F13	Q1 Output pulse width (4.0 µs)	5.5	7	14	16	18	19	3	3.72	4.28		
F14	$\overline{Q1}$	4.5	7	14	16	18	19	5				
F15	Q2	4.5	28	35	22	26	27	37				
F16	$\overline{Q2}$	5.5	28	35	22	26	27	33				

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* TABLE VI. Output pulse width test - Continued.

Test condition	Parameter	V_{CC}	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F17	Q1 Output pulse width (5.0 µs)	4.5	12	14	16	18	19	3	4.65	5.35	µs	9,10
F18	$\overline{Q1}$	5.5	12	14	16	18	19	5				
F19	Q2	5.5	24	28	22	26	27	37				
F20	$\overline{Q2}$	4.5	24	28	22	26	27	33				
F21	Q1 Output pulse width (6.0 µs)	4.5	7	12	16	18	19	3	5.58	6.42		
F22	$\overline{Q1}$	5.5	7	12	16	18	19	5				
F23	Q2	5.5	24	35	22	26	27	37				
F24	$\overline{Q2}$	4.5	24	35	22	26	27	33				
F25	Q1 Output pulse width (7.0 µs)	4.5	16	18	Open	Open	19	3	6.51	7.49		
F26	$\overline{Q1}$	5.5	16	18			19	5				
F27	Q2	5.5	22	26			27	37				
F28	$\overline{Q2}$	4.5	22	26			27	33				
F29	Q1 Output pulse width (8.0 µs)	5.5	7	18			19	3	7.44	8.56		
F30	$\overline{Q1}$	4.5	7	18			19	5				
F31	Q2	4.5	22	35			27	37				
F32	$\overline{Q2}$	5.5	22	35			27	33				

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* TABLE VI. Output pulse width test - Continued.

Test condition	Parameter	V_{CC}	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F33	Q1 Output pulse width (9.0 µs)	4.5	12	18	Open	Open	19	3	8.37	9.63	µs	9,10
	$\overline{Q1}$	5.5	12	18			19	5				
F34	Q2	5.5	22	24			27	37				
F35	$\overline{Q2}$	4.5	22	24			27	33				
F36	Q1 Output pulse width (10.0 µs)	4.5	14	18	7	12	19	3	9.3	10.7		
F37	$\overline{Q1}$	5.5	14	18	7	12	19	5				
F38	Q2	5.5	24	35	22	28	27	37				
F39	$\overline{Q2}$	4.5	24	35	22	28	27	33				
F40	Q1 Output pulse width (11.0 µs)	4.5	14	18	Open	Open	19	3	10.23	11.77		
F41	$\overline{Q1}$	5.5	14	18			19	5				
F42	Q2	5.5	22	28			27	37				
F43	$\overline{Q2}$	4.5	22	28			27	33				
F44	Q1 Output pulse width (12.0 µs)	4.5	7	14			19	3	11.16	12.84		
F45	$\overline{Q1}$	5.5	7	14			19	5				
F46	Q2	5.5	28	35			27	37				
F47	$\overline{Q2}$	4.5	28	35			27	33				
F48												

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* TABLE VI. Output pulse width test - Continued.

Test condition	Parameter	V_{CC}	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F49	Q1 Output pulse width (13.0 µs)	4.5	12	14	Open	Open	19	3	12.09	13.91	µs	9,10
F50	$\overline{Q1}$	5.5	12	14			19	5				
F51	Q2	5.5	24	28			27	37				
F52	$\overline{Q2}$	4.5	24	28			27	33				
F53	Q1 Output pulse width (14.0 µs)	4.5	7	12			19	3	13.02	14.98		
F54	$\overline{Q1}$	5.5	7	12			19	5				
F55	Q2	5.5	24	35			27	37				
F56	$\overline{Q2}$	4.5	24	35			27	33				
F57	Q1 Output pulse width (15.0 µs)	5.5	Open	Open			19	3	13.95	16.05		
F58	$\overline{Q1}$	4.5					19	5				
F59	Q2	4.5					27	37				
F60	$\overline{Q2}$	5.5					27	33				

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* TABLE VII. Transition time (t_{THL} and t_{TLH}) test.

Test condition	Parameter		Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
			E to pin	F to pin	Min	Max		
G1	Q1	Transition time (t_{THL})	19	3	3	40	ns	9,10
G2	$\overline{Q1}$		19	5				
G3	Q2		27	37				
G4	$\overline{Q2}$		27	33				
G5	$\overline{Q2}$		27	33				
G6	Q1	Transition time (t_{TLH})	19	3	10	250		
G7	$\overline{Q1}$		19	5				
G8	Q2		27	37				
G9	$\overline{Q2}$		27	33				
G10	Q1		19	3				

1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions except test conditions G5 and G10.

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* TABLE VIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Q1 Propagation delay time (t_{PLH})	19	3	10	170	ns	9,10
H2	$\overline{Q1}$ Propagation delay time (t_{PHL})	19	5	7			
H3	Q2 Propagation delay time (t_{PLH})	27	37	10			
H4	$\overline{Q2}$ Propagation delay time (t_{PHL})	27	33	7			
H5	Q1 Propagation delay time (t_{PLH})	19	3	10			
H6	$\overline{Q2}$ Propagation delay time (t_{PHL})	27	33	7			

1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions, except test conditions H5 and H6.

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* TABLE IX. Output pulse amplitude (low-level and high-level) test.

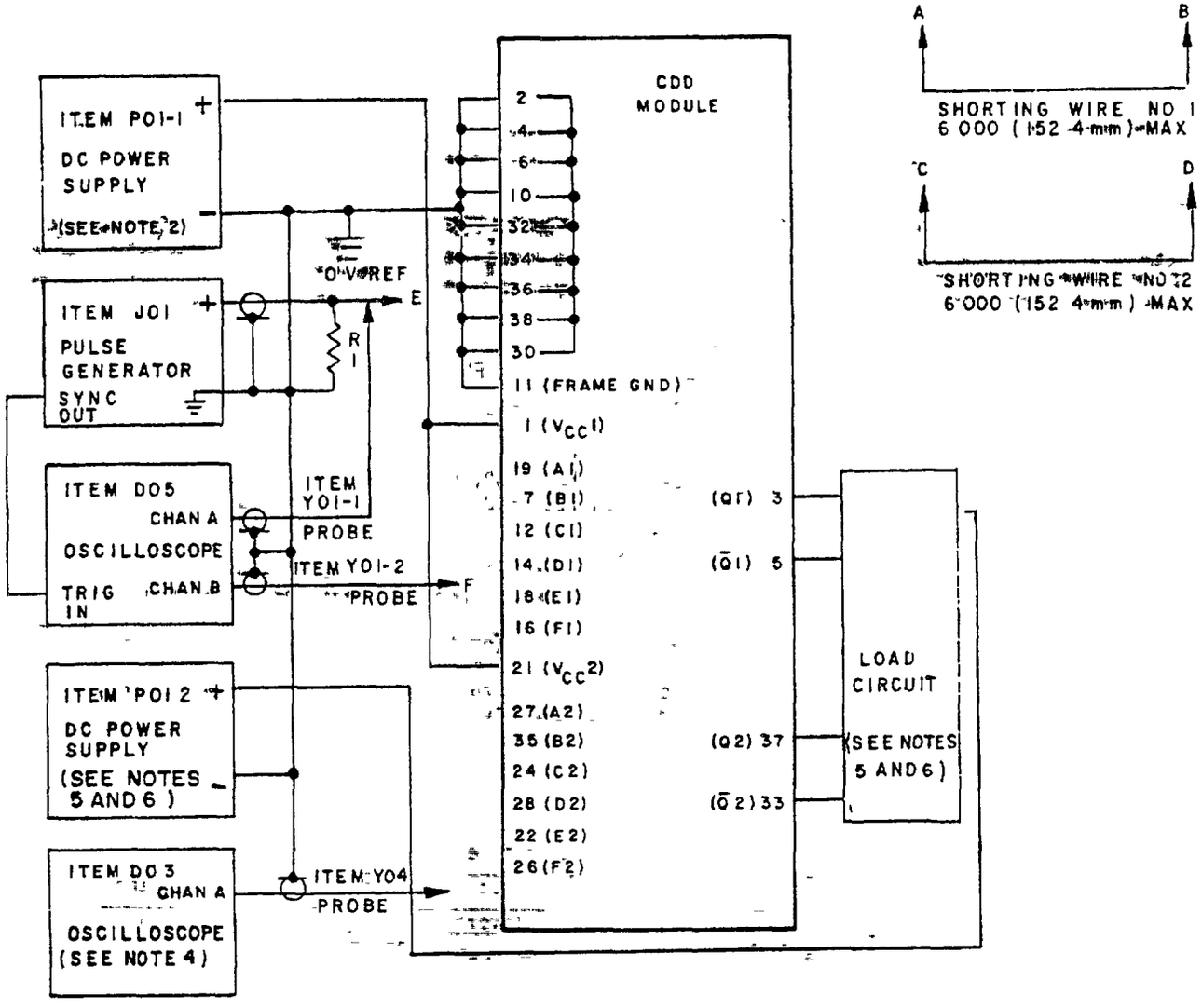
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
J1	Q1 Output pulse amplitude, low level	19	3	0	400	mV	9,10
J2	$\overline{Q1}$	19	5	↓	↓	↓	↓
J3	Q2	27	37				
J4	$\overline{Q2}$	27	33				
J5	Q1 Output pulse amplitude, high level	19	3	3.8	5.5	V	
J6	$\overline{Q1}$	19	5	↓	↓	↓	
J7	Q2	27	37				
J8	$\overline{Q2}$	27	33				

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* TABLE X. Output pulse (overshoot, undershoot, preshoot) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
K1	Q1 Output pulse, overshoot	19	3	0	500	mV	9,10
K2	$\overline{Q1}$	19	5				
K3	Q2	27	37				
K4	$\overline{Q2}$	27	33				
K5	Q1 Output pulse, undershoot	19	3				
K6	$\overline{Q1}$	19	5				
K7	Q2	27	37				
K8	$\overline{Q2}$	27	33				
K9	Q1 Output pulse, preshoot	19	3				
K10	$\overline{Q1}$	19	5				
K11	Q2	27	37				
K12	$\overline{Q2}$	27	33				

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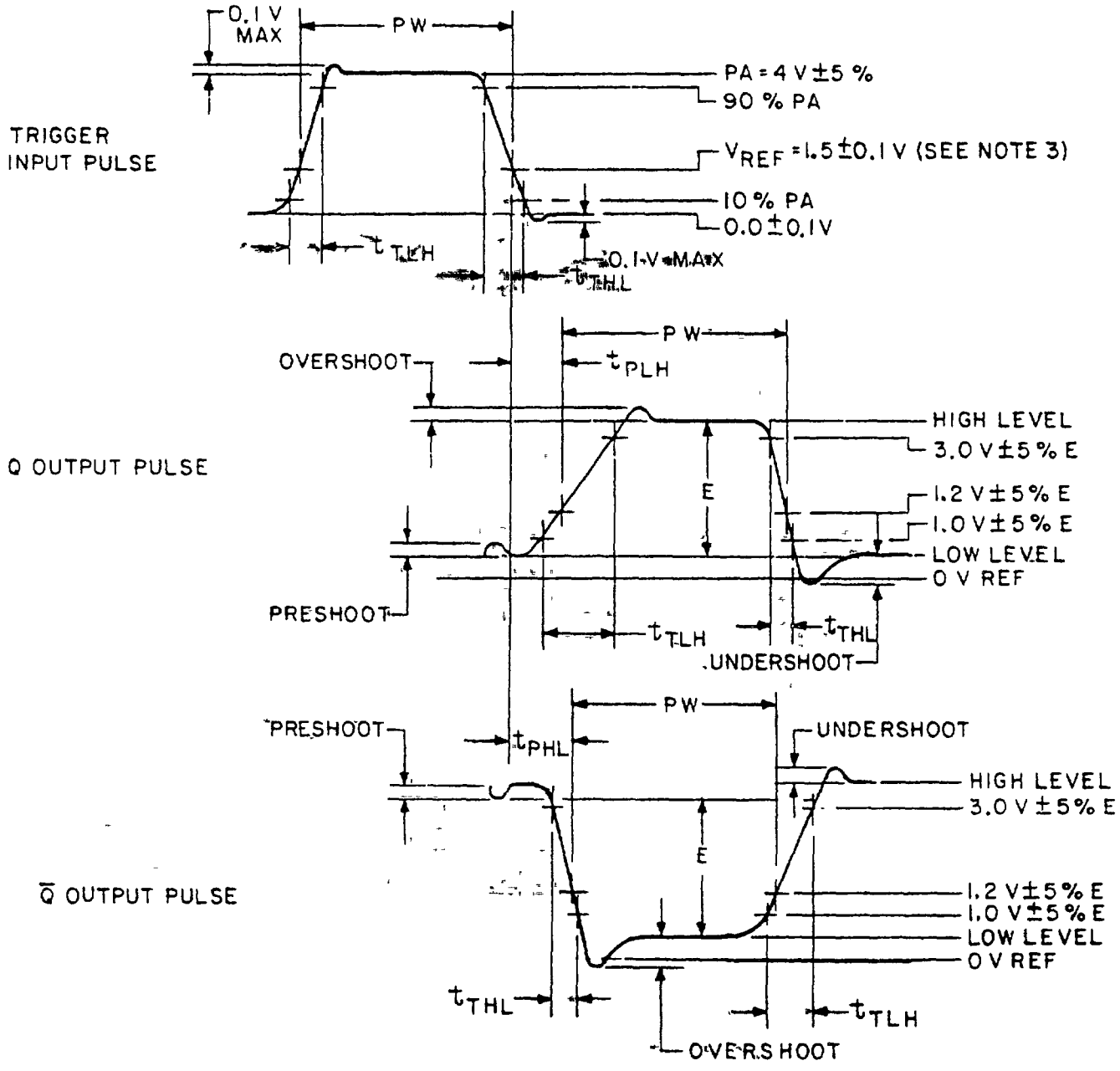
* FIGURE 9. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), and output pulse (overshoot, undershoot, preshoot) test circuit for tables VI through X.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Set item P01-1 voltage as follows:
   V_{CC} = 5.5 V ±1% for test conditions F1, F4, F5, F8, F10, F11, F13, F16, F18, F19, F22, F23, F26, F27, F29, F32, F34, F35, F38, F39, F42, F43, F46, F47, F50, F51, F54, F55, F57 and F60.
   = 4.5 V ±1% for test conditions F2, F3, F6, F7, F9, F12, F14, F15, F17, F20, F21, F24, F25, F28, F30, F31, F33, F36, F37, F40, F41, F44, F45, F48, F49, F52, F53, F56, F58, F59, G1-G10, H1-H6, J1-J8 and K1-K12.
3. For test conditions G1-G10, H1-H6, J1-J8 and K1-K12, connect item Y01-2 as close as physically possible to the module pin under test.
4. For test conditions J1-J8 use item D03 with item Y04; measurements must have 5 mV/mm resolution.
5. Use load circuit as specified on figure 3 for test conditions F1-F60 and set item P01-2 to 5.5 V ±1%.
6. Use load circuit as specified on figure 11 for test conditions G1-G10, H1-H6, J1-J8 and K1-K12 and set item P01-2 to 2.5 V ±1%.

* FIGURE 9. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), and output pulse (overshoot, undershoot, preshoot) test circuit for tables VI through X - Continued.

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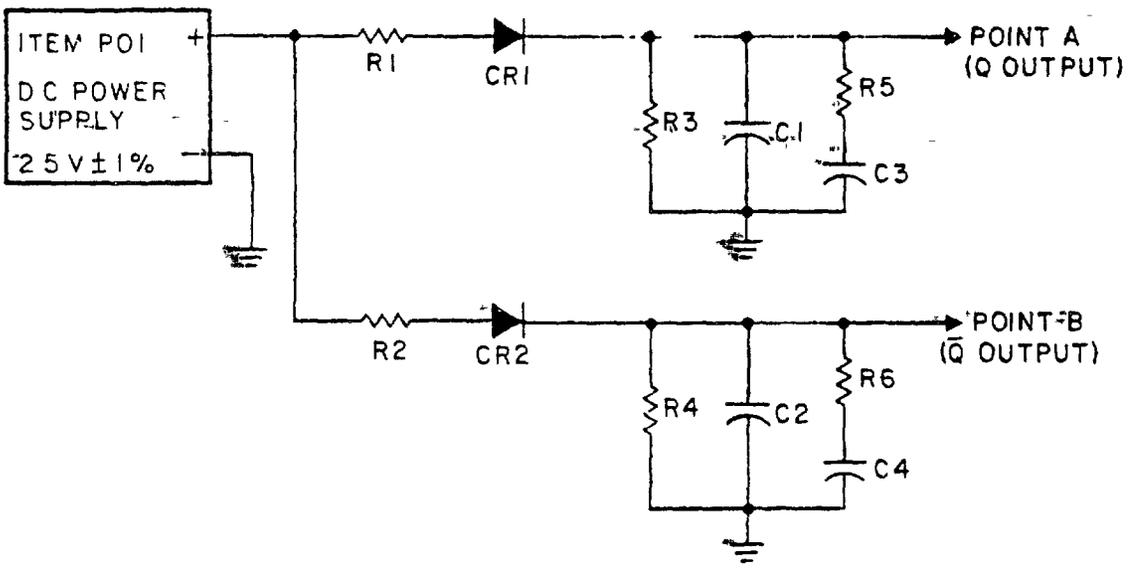
* FIGURE 10. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level) and high-level), output pulse (overshoot, undershoot, preshoot), and input trigger sensitivity and immunity test circuit waveforms for tables VI through X and XII.

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NOTES:
1. Input pulse parameters for test conditions F1-F8, F13-F16, F29-F32, F57-F60, G1-G10, H1-H6, J1-J8 and K1-K12:
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PRP = 30 µs ±10%.
2. Input pulse parameters for test conditions F9-F12, F17-F28, and F33-F56:
   PW (1.5 V level) = 100 µs ±10%.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PRP = 200 µs ±10%.
3. Input pulse width is measured at 50% PA for test conditions M1 through M12.
4. Several perturbations which may occur on the output waveform are shown (preshoot, overshoot, undershoot). Perturbations may occur at other locations on the output waveforms, but they must conform to the following requirements: No negative perturbations on the high level or positive perturbations on the low level shall exceed 0.5 V amplitude.

* FIGURE 10. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot, preshoot), and input trigger sensitivity and immunity test circuit waveforms for tables VI through X and XII - Continued.

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NOTES:
1. R3 = R4 = 18.7 kΩ ±1%, 1/8 W.
2. R5 = R6 = 100Ω ±1%, 1/8 W.
3. C1 = C2 = 33 pF ±10%.
4. C3 = C4 = 330 pF ±10%.
5. CR1 = CR2 = 1N4150, or equivalent.
6. R1 = Select value for a current flow of 35.2 mA ±1% into point A when point A is held at 0.4 V.
7. R2 = Select value for a current flow of 22.4 mA ±1% into point B when point B is held at 0.4 V.
8. This load shall be applied to each module circuit Q and $\overline{Q}$ output.

* FIGURE 11. CDD logic load circuit.

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* TABLE XI. Output pulse width (recovery time) test.

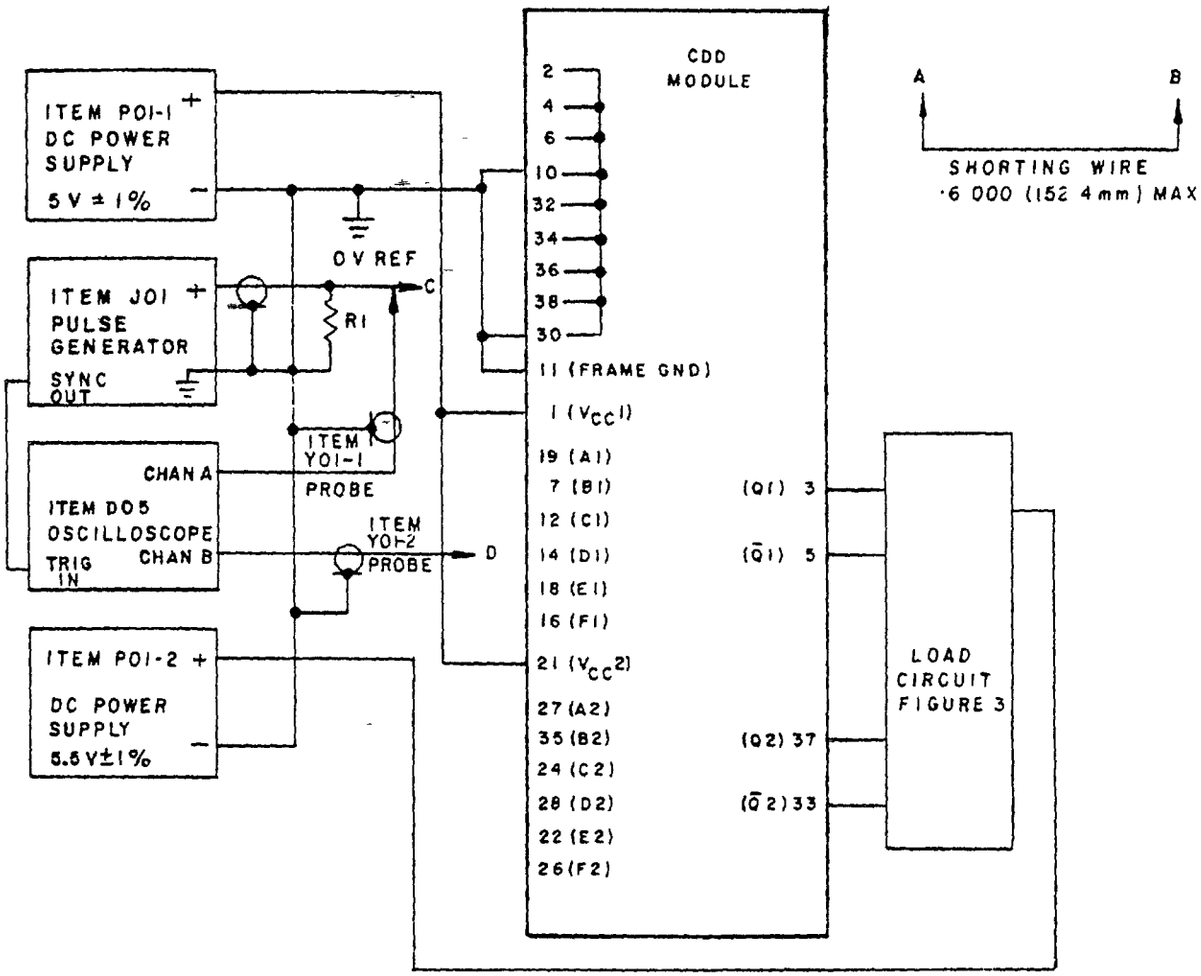
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	Q1 Output pulse width (recovery time)	12	16	19	3	0.93	1.07	µs	12
L2	Q1 ↓	Open	Open	19	3	13.95	16.05	↓	↓
L3	Q2	24	26	27	37	0.93	1.07		
L4	Q2	Open	Open	27	37	13.95	16.05		

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* TABLE XII. Input trigger sensitivity and immunity test.

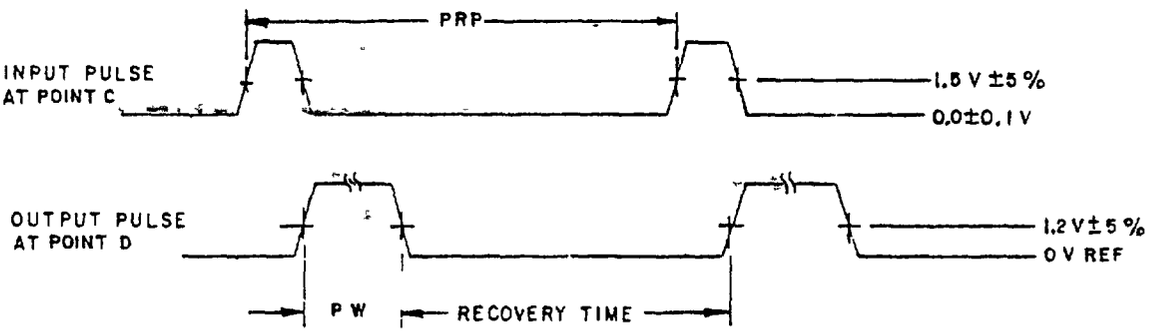
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	Q1 Input trigger sensitivity	12	16	19	3	0.93	1.07	µs	10,12
M2	$\overline{Q1}$	12	16	19	5				
M3	Q1	24	26	27	37				
M4	$\overline{Q2}$	24	26	27	33				
M5	Q1 Input immunity (20 ns)	12	16	19	3	0	5	ns	
M6	$\overline{Q1}$	12	16	19	5				
M7	Q2	24	26	27	37				
M8	$\overline{Q2}$	24	26	27	33				
M9	Q1 Input immunity (0.95 V)	12	16	19	3				
M10	$\overline{Q1}$	12	16	19	5				
M11	Q2	24	26	27	37				
M12	$\overline{Q2}$	24	26	27	33				

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* FIGURE 12. Output pulse width (recovery time), input trigger sensitivity and immunity test circuit for tables XI and XII.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. For test conditions L1 and L3 item J01 output shall be adjusted as follows:
   PA = 4 V ±5% with reference to 0 V.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 5 µs ±5%.
3. For test conditions L2 and L4 item J01 output shall be adjusted as follows:
   PA = 4 V ±5% with reference to 0 V.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 19 µs ±5%.
4. For test conditions M1 through M4 item J01 output shall be adjusted as follows:
   PA = 3.2 V ±5% with reference to 0 V.
   PW (50% level) = 135 ns ±5%.
   $t_{TLH} = t_{THL}$ = 40 ns ±5%.
   PRP = 100 kHz ±10%.
5. For test conditions M5 through M8 item J01 output shall be adjusted as follows:
   PA = 5.0 V ±5% with reference to 0 V.
   PW (50% level) = 20 ns ±5%.
   $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   PRP = 100 kHz ±10%.
6. For test conditions M9 through M12 item J01 output shall be adjusted as follows:
   PA = 0.95 V ±5% with reference to 0 V.
   PW (50% level) = 1000 ns ±5%.
   $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   PRP = 100 kHz ±10%.

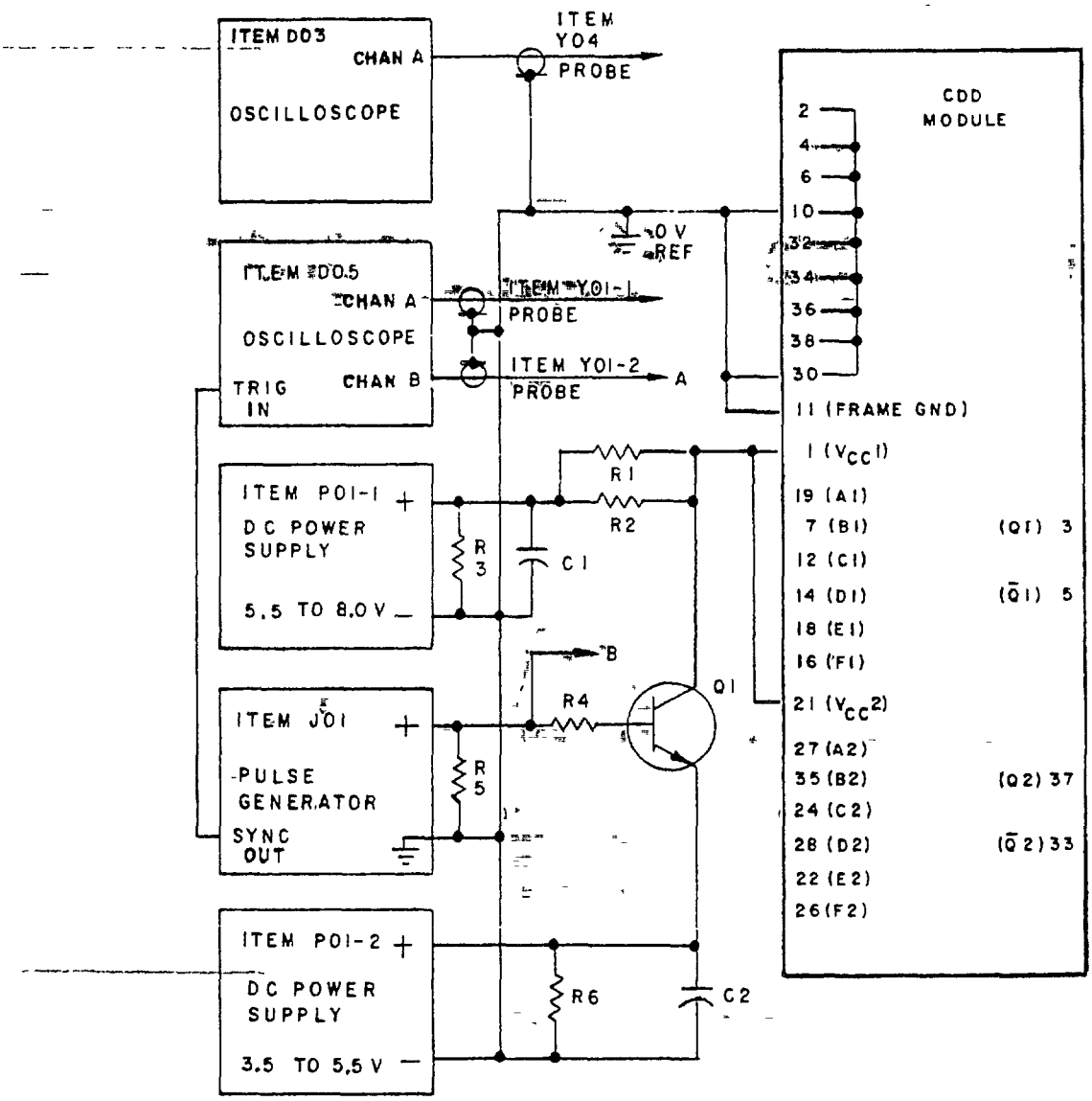
* FIGURE 12. Output pulse width (recovery time), input trigger sensitivity and immunity test circuit for tables XI and XII - Continued.

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* TABLE XIII. Power supply transient (rise time and false trigger) test.

Test condition	Parameter		Connect A to pin,	Initial limits 25°±5°C ambient Min	Max	Unit	Test figure
N1	$\overline{Q1}$	Power supply transient rise time	5	9	53	µs	13
N2	$\overline{Q2}$	↓	33	9	53	↓	↓
N3	$\overline{Q1}$	↓	5	1.5	40.0	↓	↓
N4	$\overline{Q2}$	↓	33	1.5	40.0	↓	↓
N5	$\overline{Q1}$	Power supply transient false trigger	5	2.0	5.5	V	↓
N6	$\overline{Q2}$	↓	33	↓	↓	↓	↓
N7	$\overline{Q1}$	↓	5	↓	↓	↓	14
N8	$\overline{Q2}$	↓	33	↓	↓	↓	14

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* FIGURE 13. Power supply transient (rise time and false trigger) test circuit for table XIII.

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NOTES:
1. R1 = R2 = 10Ω ±1%, 1 W.
2. R3 = R6 = 18.2Ω ±1%, 3 W.
3. R4 = R5 = 51.1Ω ±1%, 1/2 W.
4. C1 = C2 = 1.0 µF ±10%.
5. Q1 = 2N3108A or equivalent.

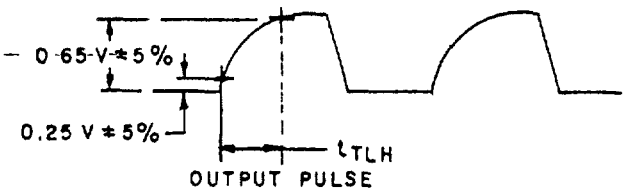


6. Using item D03, adjust item J01 to obtain the preceding waveform at point B with the output of item J01 disconnected from R4.
$t_{TLH} = t_{THL}$ = 50 ns ±20%.
PW (at 50% amplitude) = 400 µs ±10%.
Duty cycle = 50%.



VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions N1 through N6, connect item J01 to R4; connect item Y04 to module pin 1; and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5 V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9 V ±5% as shown above.

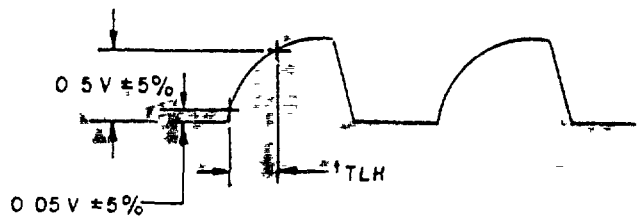


8. For test conditions N1 and N2, the output pulse rise time shall be measured as shown above and shall conform to the requirements of table XIII using item D05 and item Y01-2 (with 1X attenuator and AC coupler).

* FIGURE 13 Power supply transient (rise time and false trigger) test circuit for table XIII - Continued.

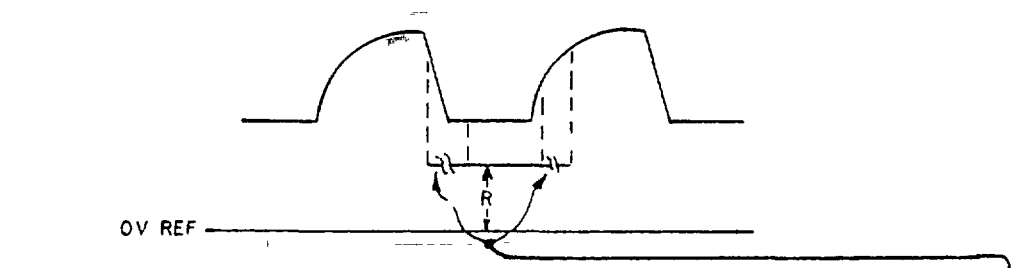
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9. For test conditions N3 and N4, the output pulse rise time shall be measured as shown below and shall conform to the requirements of table XIII using item D05 and item Y01-2 (with 1X attenuator and AC coupler).



OUTPUT PULSE

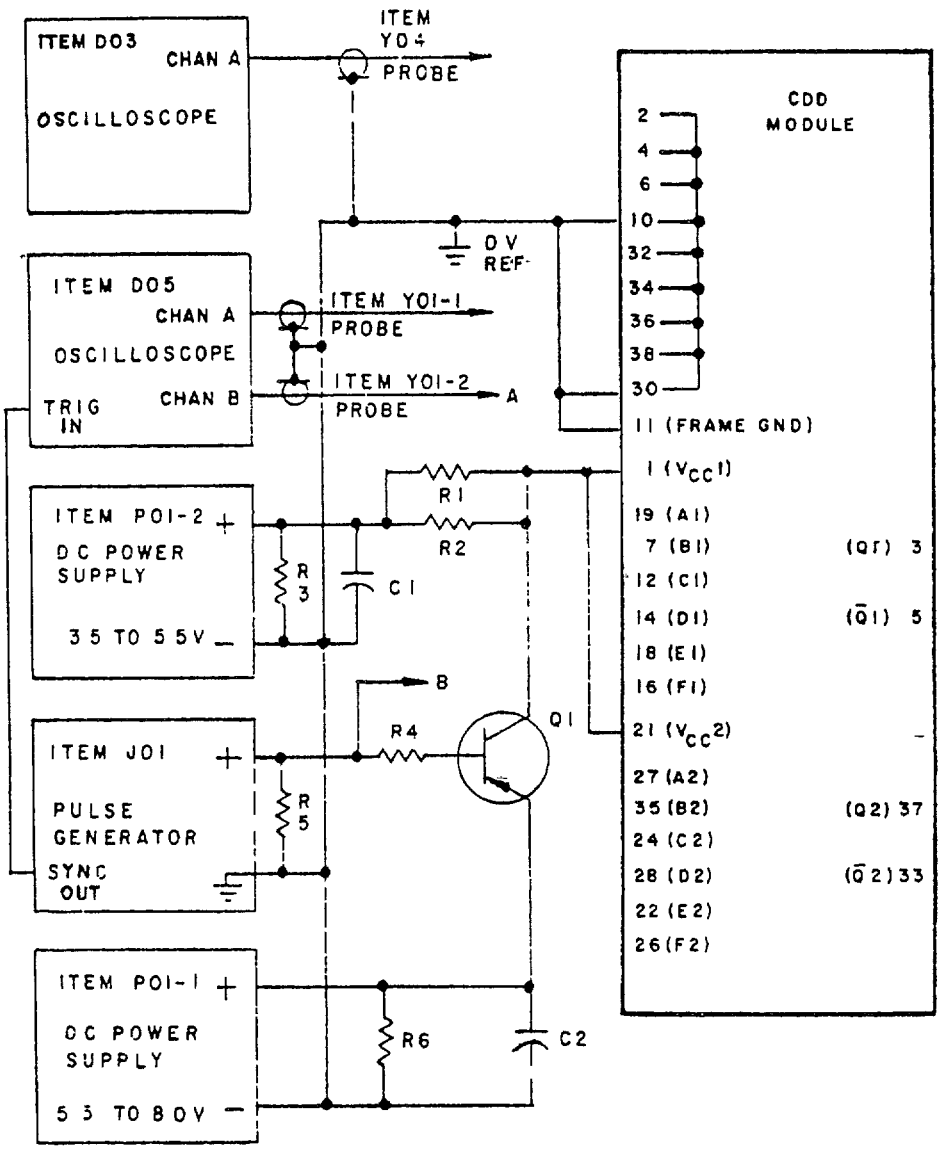
10. For test conditions N5 and N6, the output pulse measurement (R) shall conform to the requirements of table XIII using item D03. The value R is defined as the minimum voltage of the waveform.



OUTPUT PULSE

* FIGURE 13. Power supply transient (rise time and false trigger) test circuit for table XIII - Continued

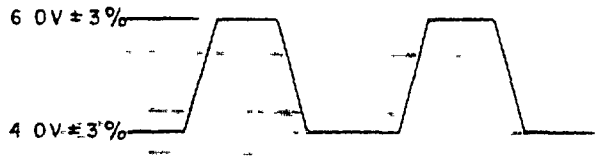
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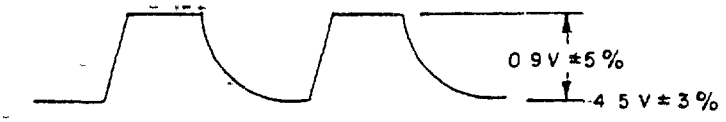
* FIGURE 14. Power supply transient false trigger test circuit for table XIII.

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NOTES:
1. R1 = R2 = 10Ω ±1%, 1 W.
2. R3 = R6 = 18.2Ω ±1%, 3 W.
3. R4 = R5 = 51.1Ω ±1%, 1/2 W.
4. C1 = C2 = 1.0 µF ±10%, 50 Vdc.
5. Q1 = 2N4029 or equivalent.

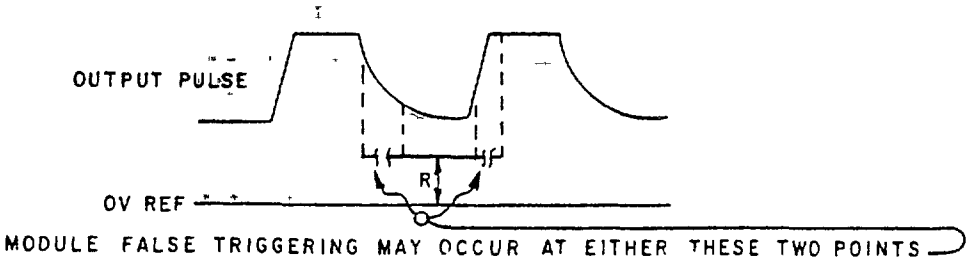


6. Using item D03, adjust item J01 to obtain the preceding waveform at point B with the output of item J01 disconnected from R4.
$t_{TLH} = t_{THL}$ = 50 ns ±10%.
PW (at 50% amplitude) = 400 µs ±10%.
Duty cycle = 50%.



VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions N7 and N8, connect item J01 to R4, connect item Y04 to module pin 1, and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5 V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9 V ±5% as shown above.



8. For test conditions N7 and N8, the output pulse measurement (R) shall conform to the requirements of table XIII using item D03. The value R is defined as the minimum voltage as shown on the above waveform.

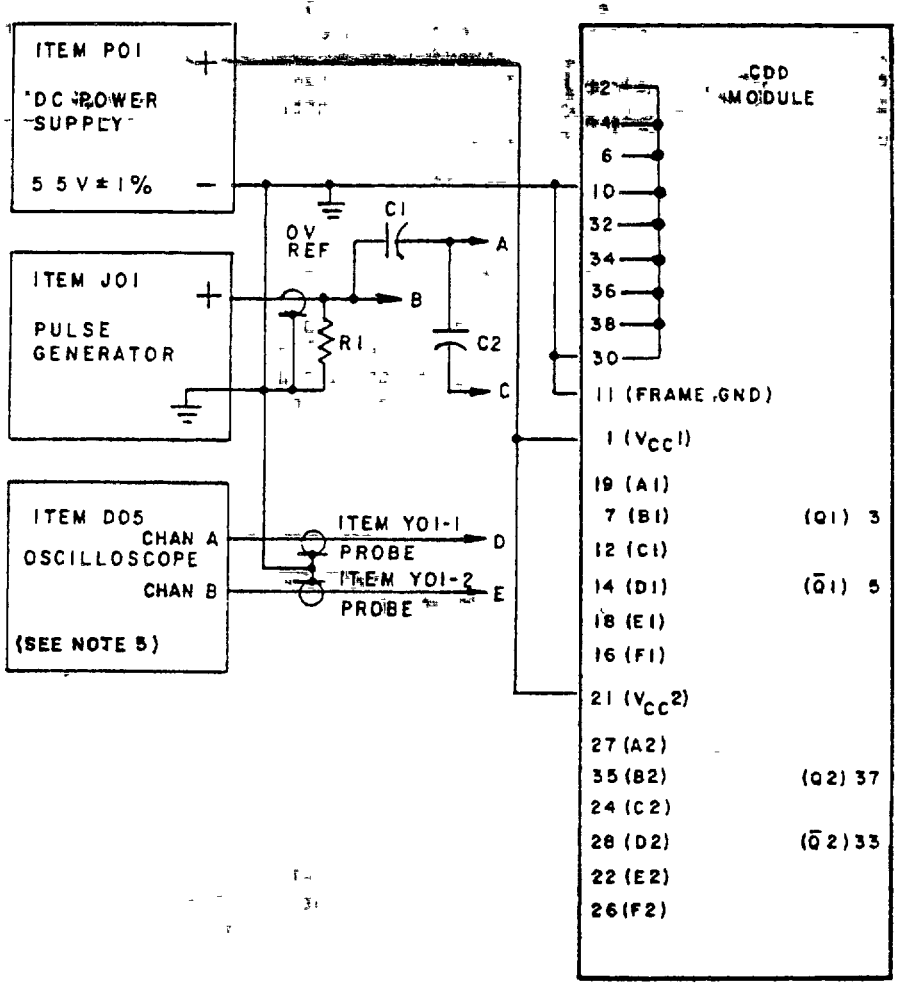
* FIGURE 14. Power supply transient false trigger test circuit for table XIII - Continued.

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* TABLE XIV. Crosstalk test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P1	Q1	Crosstalk	16	27	33	27	3	0	5	ns	15
P2	Q1	↓	↓	↓	↓	33	3	↓	↓	↓	↓
P3	Q1	↓	↓	↓	↓	33	3	↓	↓	↓	↓
P4	$\overline{Q1}$	↓	↓	↓	↓	27	5	↓	↓	↓	↓
P5	$\overline{Q1}$	↓	↓	↓	↓	33	5	↓	↓	↓	↓
P6	$\overline{Q1}$	↓	↓	↓	↓	33	5	↓	↓	↓	↓
P7	Q2	↓	26	19	5	19	37	↓	↓	↓	↓
P8	Q2	↓	↓	↓	↓	5	37	↓	↓	↓	↓
P9	Q2	↓	↓	↓	↓	5	37	↓	↓	↓	↓
P10	$\overline{Q2}$	↓	↓	↓	↓	19	33	↓	↓	↓	↓
P11	$\overline{Q2}$	↓	↓	↓	↓	5	33	↓	↓	↓	↓
P12	$\overline{Q2}$	↓	↓	↓	↓	5	33	↓	↓	↓	↓

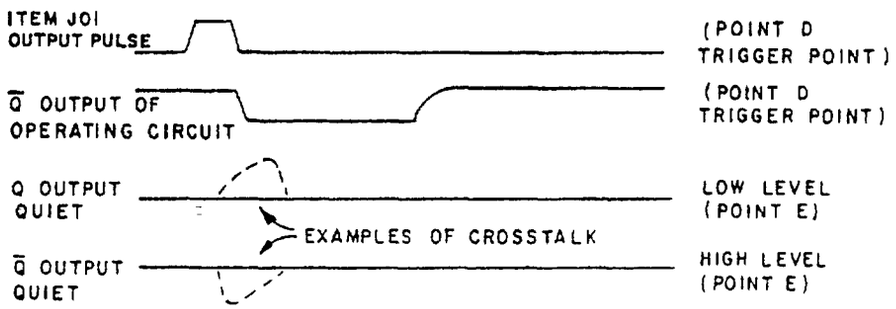
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* FIGURE 15. Crosstalk test circuit for table XIV.

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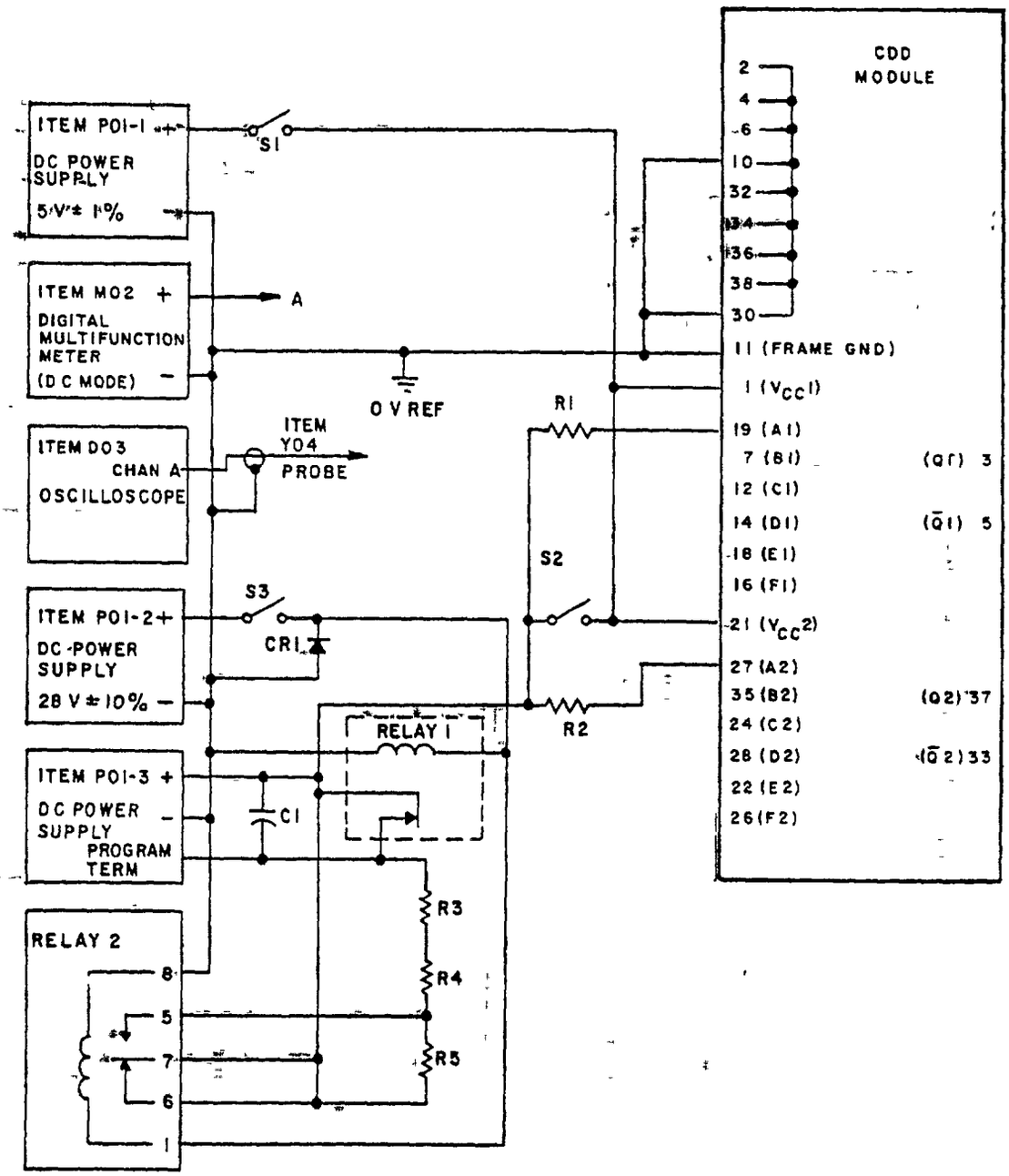
NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. C1 = 4 pF maximum.
3. C2 = 5 ±1 pF.
4. The capacitances are measured using a capacitance bridge item C02 at the test fixture pins with the module removed, at a frequency of 1 MHz ±10%, and will include stray wiring capacitance but not oscilloscope probe capacitance. Glass trim capacitors may be added to conform to the required values.
5. Set item D05 internal trigger slope to negative for test conditions P1, P3, P4, P6, P7, P9, P10 and P12 and to positive for test conditions P2, P5, P8 and P11.
6. Using item D05, adjust the output of item J01 to obtain the following input pulse:
   PA = 4 V ±5%.
   Pulse width (2 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ns ±10%.
   PRP = 100 µs ±10%.
7. This test checks the effect of the operating circuit on the nonoperating circuit. The test requires the detection and examination of output pulses, if present, on the nonoperating circuit. To examine the output waveforms under test, item D05 shall be internally triggered using the signal on Channel A (item Y01-1). Two different signals shall be used for triggering as part of each test condition: The output pulse of item J01 and the $\overline{Q}$ output of the operating circuit. Table XIV lists the three signal trigger points used.



For each test condition, the maximum pulse width detected at point E, measured at 1.0 V above the low level on the Q output and at 3.0 V below the high level on the $\overline{Q}$ output, shall not exceed the limits of table XIV.

* FIGURE 15. Crosstalk test circuit for table XIV - Continued.

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* FIGURE 16. Zener applied potential and trigger voltage high test circuit for table XV.

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* TABLE XV. Zener applied potential and trigger voltage high test.

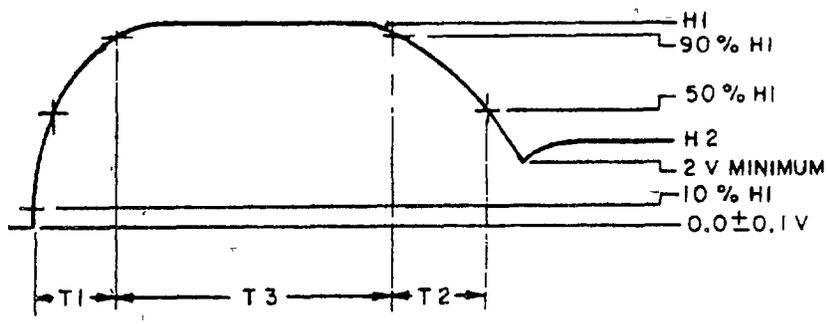
Test Condition	Parameter	S1	S2	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q1	$\overline{Q1}$ Zener applied potential	Open	Closed	5	3.8	5.5	V	16
Q2	$\overline{Q2}$ Zener applied potential	Open	Closed	33	3.8	5.5	V	
Q3	Q1 Trigger voltage high	Closed	Open	3	0	400	mV	
Q4	Q2 Trigger voltage high	Closed	Open	37	0	400	mV	

* TABLE XVI. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R1	Q1 V_{OUT} (low loaded)	3	0.0	1.1	V	17
R2	Q2 V_{OUT} (low loaded)	37	0.0	1.1	V	17

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NOTES:
1. R1 = R2 = 154Ω ±1%, 1/8 W.
2. R3 = 178Ω ±1%, 1/8 W.
3. R4 = 301Ω ±1%, 1/8 W.
4. R5 = 649Ω ±1%, 1/8 W.
5. C1 = 22 µF ±10%.
6. CR1 = 1N647 or equivalent.
7. S1 = S2 = S3 = SPST switch.
8. Relay 1 = GE model 3SAV1266AZ or equivalent.
9. Relay 2 = Agastat model 2112 or equivalent.
10. Resistors R3, R4, R5, and capacitor C1 are programming components for item P01-3 and are based on a programming constant of 100Ω per volt.
11. Program item P01-3 and using item D03 with item Y04, adjust the time delay relay to obtain the following output waveform at pin 7 of the time delay relay after switch S3 closed.

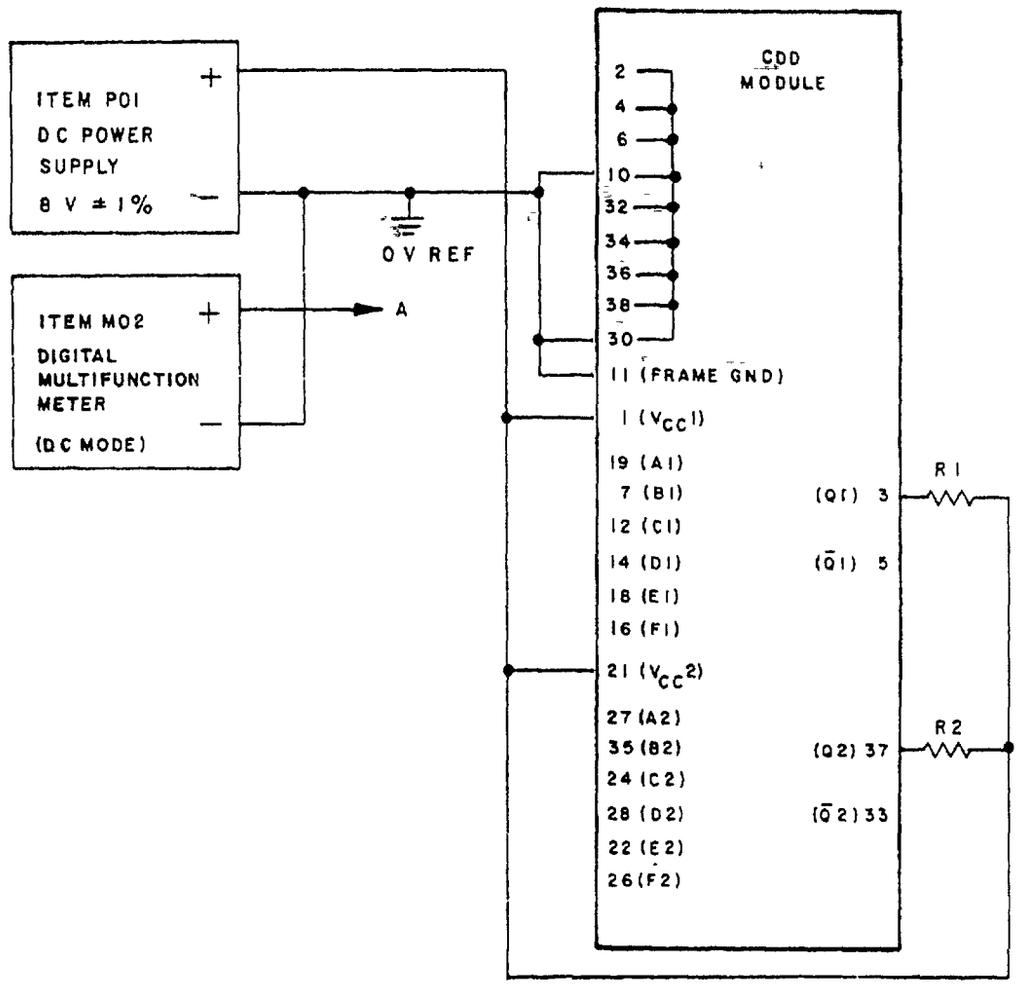


Waveform parameters:
H1 = 11.25 V ±1%.
H2 = 5.0 V ±10%.
T1 = 100 $^{+100}_{-50}$ ms.
T2 = 100 $^{+400}_{-50}$ ms.
T3 = 0.5 sec ±10%.

12. Item P01-3 should be selected to ensure that the programming terminal is floating (-).
13. Test sequence: (a) Close switch S3, (b) wait 2 seconds minimum, (c) make voltage measurements for test conditions, and (d) open switch S3.

* FIGURE 16. Zener applied potential and trigger voltage high test circuit for table XV - Continued.

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NOTES:
1. R1 = R2 = 82Ω ±5%, 3 W.
2. The measurement for each output shall be made after a time period of at least 10 seconds.

* FIGURE 17. V_{OUT} (low-loaded) test circuit for table XVI.

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